

Cloud Computing

Mock Exam 3 | Course ID: noc26_cs55

Q1. What is 'Cloudonomics' primarily concerned with?

- A) Cloud security protocols
- B) Economic implications of cloud computing
- C) Cloud network architecture
- D) Cloud software development

Source: Lecture_W1L19_Lecture_1__Cloud_Computing__Overview | Difficulty: Easy

Q2. Which of the following is an essential characteristic of cloud computing according to the NIST definition?

- A) Fixed pricing
- B) On-demand self-service
- C) Dedicated hardware
- D) Manual provisioning

Source: Lecture_W1L21_Lecture_3__Cloud_Computing__Introduction | Difficulty: Easy

Q3. Which cloud service model delivers ready-to-use software applications over the internet, typically via a web browser?

- A) IaaS
- B) PaaS
- C) SaaS
- D) XaaS

Source: Lecture_W1L21_Lecture_3__Cloud_Computing__Introduction | Difficulty: Easy

Q4. Which scaling method is described as adding more machines (nodes) to a resource pool to increase capacity, and is the primary method for achieving elasticity in the cloud?

- A) Vertical Scaling
- B) Diagonal Scaling
- C) Horizontal Scaling
- D) Radial Scaling

Source: Lecture_W1L22_Lecture_4__Cloud_Computing_Architecture | Difficulty: Medium

Q5. What is identified as the core differentiator between cloud computing and the traditional client-server model?

- A) Network speed
- B) Data storage capacity
- C) Virtualization
- D) Programming language

Source: Lecture_W1L23_Lecture_5__Cloud_Computing_Architecture__contd__ | Difficulty: Medium

Q6. Which cloud deployment model is provisioned for open use by the general public and owned/operated by a third-party provider?

- A) Private Cloud
- B) Community Cloud
- C) Hybrid Cloud
- D) Public Cloud

Source: Lecture_W2L28_Lecture_6__Cloud_Computing_Architecture__Deploym | Difficulty: Easy

Q7. In an IaaS environment, what is the primary role of the hypervisor (VMM)?

- A) Managing application data
- B) Creating and running virtual machines
- C) Developing software applications
- D) Providing network connectivity

Source: Lecture_W2L29_Lecture_7__Cloud_Computing__Virtualization | Difficulty: Medium

Q8. What is the primary purpose of XML (Extensible Markup Language)?

- A) To display data for humans
- B) To structure and transport data for machines
- C) To manage database transactions
- D) To define network protocols

Source: Lecture_W2L30_Lecture_8__Cloud_Computing__XML_Basics | Difficulty: Easy

Q9. Which XML parser is described as event-based, reading sequentially without storing the entire document in memory, making it ideal for very large XML files?

- A) DOM Parser
- B) SAX Parser
- C) XSD Parser
- D) XSLT Parser

Source: Lecture_W2L31_Lecture_9__Cloud_Computing__XML_Basics_II | Difficulty: Medium

Q10. In the context of Web Services, what does WSDL (Web Services Description Language) primarily describe?

- A) Data encryption standards
- B) A service's capabilities and how to interact with it
- C) Network routing paths
- D) User authentication methods

Source: Lecture_W2L32_Lecture_10__Cloud_Computing__Web_Services__Service | Difficulty: Easy

Q11. In the SLA hierarchy, which component is at the base, representing raw, system-level metrics like downtime or bytes in/out?

- A) Service Level Agreement (SLA)
- B) Service Level Objective (SLO)
- C) Key Performance Indicator (KPI)
- D) Cloudonomics

Source: Lecture_W3L37_Lecture_11__Service_Level_Agreement | Difficulty: Medium

Q12. According to the lecture, the cloud is more economical when which condition is met, where P is peak demand, A is average demand, and U is the utility premium?

- A) $P/A < U$
- B) $P * A > U$
- C) $P / A > U$
- D) $P + A = U$

Source: Lecture_W3L38_Lecture_12__Cloud_Economics | Difficulty: Hard

Q13. Which core feature of GFS/HDFS enables reliable storage and parallel data access at scale?

- A) Storing data in small, encrypted blocks
- B) Breaking files into large chunks and replicating them
- C) Using a single, centralized server for all data
- D) Prioritizing immediate consistency over availability

Source: Lecture_W3L39_Lecture_13__Managing_Data | Difficulty: Medium

Q14. In the MapReduce analysis of word counting, which approach was shown to be highly scalable by dividing the documents among processors?

- A) Divide the words
- B) Divide the processors
- C) Divide the documents
- D) Divide the output

Source: Lecture_W3L40_Lecture_14__Introduction_to_Map_Reduce | Difficulty: Medium

Q15. In OpenStack, which component is the primary compute engine responsible for managing the lifecycle of virtual machines?

- A) Horizon
- B) Neutron
- C) Nova
- D) Glance

Source: Lecture_W3L41_Lecture_15__Open_Stack | Difficulty: Easy

Q16. In Microsoft Azure's architectural blueprint, which computational role is designed for running background processing and distributed tasks without direct user interaction?

- A) Web Role
- B) Frontend Role
- C) Worker Role
- D) Management Role

Source: Lecture_W4L47_Lecture_17__Cloud_Computing__Case_Study_with_a_Co | Difficulty: Easy

Q17. A service has a guaranteed availability of 99.95%. Over a 30-day period, operating 12 hours/day, the total downtime was 9.417 hours. The initial cost was \$1500. If availability < 99% triggers a 25% credit, what is the effective cost?

- A) \$1500
- B) \$1125
- C) \$1350
- D) \$1000

Source: Lecture_W5L55_Lecture_21__SLA_Tutorial | Difficulty: Hard

Q18. In a cost comparison, an in-house server has a total cost of \$85.08/hr at 40% efficiency. A cloud alternative costs \$53.50/hr. At what approximate efficiency percentage would the in-house server become cheaper than the cloud alternative?

- A) 40%
- B) 70%
- C) 90%
- D) 100%

Source: Lecture_W5L56_Lecture_22__Cloud_Computing__Economics_Tutorial | Difficulty: Hard

Q19. What is the primary energy-saving strategy in Green Cloud Computing that involves packing virtual machines onto the minimum number of physical servers and powering down idle ones?

- A) Live Migration
- B) VM Image Minimization
- C) Workload Consolidation
- D) Carbon-Aware Scheduling

Source: Lecture_W5L58_Lecture_24__Resource_Management_I | Difficulty: Medium

Q20. Which resource management concept involves the negotiation of resources through an intermediary agent to ensure they are available at the right time and for the right price?

- A) Resource Provisioning
- B) Resource Allocation
- C) Resource Brokering
- D) Resource Scheduling

Source: Lecture_W5L59_Lecture_25_Resource_Management_II | Difficulty: Medium

Q21. Which of the following is one of the three foundational pillars of information security in the CIA Triad?

- A) Complexity
- B) Cost
- C) Confidentiality
- D) Compliance

Source: Lecture_W6L64_Lecture_26_Cloud_Computing_Security_I | Difficulty: Easy

Q22. Why is a compromised hypervisor considered the 'key to the kingdom' in cloud security?

- A) It only affects a single VM
- B) It allows an attacker to control all VMs on the physical machine
- C) It only impacts network traffic
- D) It only affects data storage

Source: Lecture_W6L65_Lecture_27_Cloud_Computing_Security_II | Difficulty: Hard

Q23. In the 'Hey, You, Get Off of My Cloud' case study, what was the primary goal of the placement phase of a co-residence attack?

- A) To steal network bandwidth
- B) To intentionally place a malicious VM on the same physical machine as a targeted VM
- C) To overload the cloud provider's network
- D) To encrypt the victim's data

Source: Lecture_W6L66_Lecture_28_Cloud_Computing_Security_III | Difficulty: Hard

Q24. In collaborative SaaS clouds, what is a 'Cyclic Inheritance Conflict'?

- A) A conflict where a user cannot access their own data
- B) A vulnerability where a user gains unauthorized permissions by traversing a cycle of roles across collaborating domains
- C) A conflict in data storage locations
- D) A conflict in billing models

Source: Lecture_W6L67_Lecture_29_Cloud_Computing_Security_Issues_In_Co | Difficulty: Hard

Q25. What is the primary role of an intelligent Cloud Broker in the cloud marketplace?

- A) To develop new cloud services
- B) To manage physical data centers
- C) To help customers select the optimal provider and manage service lifecycle
- D) To provide raw compute resources

Source: Lecture_W6L68_Lecture_30_Cloud_Computing_Broker_for_Cloud_Mark | Difficulty: Medium

Q26. What is the primary benefit of 'Computation Offloading' in Mobile Cloud Computing (MCC)?

- A) Increasing device storage
- B) Saving battery and enabling execution of resource-intensive applications
- C) Improving screen resolution
- D) Reducing network latency

Source: Lecture_W7L73_Lecture_31_Mobile_Cloud_Computing_I | Difficulty: Medium

Q27. According to the lecture, offloading is beneficial when which condition is met regarding computation (C), data transfer (D), mobile speed (M), and cloud speed ($S=F*M$)?

- A) C is small and D is large
- B) C is large and D is small
- C) M is greater than S
- D) F is less than 1

Source: Lecture_W7L74_Lecture_32__Mobile_Cloud_Computing__II | Difficulty: Hard

Q28. Fog computing is described as an 'extension' of cloud computing, not a replacement. What is its primary advantage for edge-heavy applications?

- A) Provides massive storage
- B) Reduces latency and network load by enabling local processing
- C) Offers lower cost for global analytics
- D) Simplifies core network infrastructure

Source: Lecture_W7L75_Lecture_33__Fog_Computing_I | Difficulty: Medium

Q29. In Docker terminology, what is an 'Image'?

- A) A running instance of an application
- B) A blueprint or template for a container
- C) A virtual machine
- D) A network configuration file

Source: Lecture_W8L82_Lecture_36__Introduction_to_DOCKER_Container | Difficulty: Easy

Q30. According to the lecture, what critical insight is revealed by the energy breakdown in data centers, regarding where the majority of power is used?

- A) Most power is used by networking equipment
- B) The majority of power is *not* used for computation itself
- C) Power distribution consumes the least energy
- D) Servers consume 100% of their peak power even when idle

Source: Lecture_W8L83_Lecture_37__Green_Cloud | Difficulty: Hard

Answer Key

1: B	2: B	3: C	4: C	5: C
6: D	7: B	8: B	9: B	10: B
11: C	12: C	13: B	14: C	15: C
16: C	17: B	18: C	19: C	20: C
21: C	22: B	23: B	24: B	25: C
26: B	27: B	28: B	29: B	30: B